Autonomous Robot for Assessing Accessibility [128]

Steve Greedy (steve.greedy@nottingham.ac.uk)

Brief Description of the Project
Design and development of an autonomous robot for assessing accessibility in buildings for wheelchair users

Available to students on the following degrees

Electrical & Electronic Eng.	Electrical Engineering	Electronic Engineering	Electronic & Computing Eng.
Yes	No	Yes	Yes

Key areas
Robotics
AI

Sub Areas
Object detection algorithms
environmental mapping

Distinctive Skills to be gained.
Practical robot design, embedded systems/sensors, AI for object detection.

Nature of the project

- Programming: 50% of the work
- Construction: 25% of the work
- Simulation/Modelling: None
- Practical lab work: 25% of the work

Modules appropriate for this work
All core EEE modules

Extended description of the project
This project involves the development of an autonomous robot capable of assessing accessibility in public buildings or other spaces for wheelchair users.

Potential aims and objectives of the project
The project aims to develop an autonomous or semi-autonomous robot capable of traversing an indoor space and use on board sensor systems to gather data that can be used to assess accessibilty for wheelchair users.

- Design and fabrication of a compact robotic platform that can move around an indoor environment within the spatial constraints of a standard wheelchair
- Equip the robot with systems to measure data on the environment to determine whether spaces are accessible to a wheelchair
- Develop autonomous or semi-autonomous capabiities to allow the robot to move and identify potential barriers to accessibility.
- Gather data to generate a report on accessibility issues that may be faced.
- Provide real-time data via a web interface on potential accessibility issues